Math 242, S19 **Name:** Answer

Quiz 9

__Instructions:__ **There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (10 points) Find the general solution of the given nonhomogeneous differential equation:

$$y'' - y' - 2y = 2x^2 + 3.$$

Solution: step 1. Find the complimentary solution $y_c(x)$

Characteristic equation

$$r^2 - r - 2 = 0$$

$$(r-2)(r+1) = 0 \Rightarrow r_1 = 2, \ r_2 = -1$$

Thus $y_c(x) = c_1 e^{2x} + c_2 e^{-x}$

Step 2. Find a particular solution $y_p(x)$

$f(x) = 2x^2+3$, $f'(x) = 4x$, $f''(x) = 4$, $f'''(x) = 0$

Thus we take the trial function

$$y_p(x) = Ax^2 + Bx + C$$

$y_p'(x) = 2Ax + B$, $y_p''(x) = 2A$

Thus

$$\begin{aligned} y_p'' - y_p' - 2y_p &= (2A) - (2Ax+B) - 2(Ax^2+Bx+C) \\ &= (-2A)x^2 + (-2A-2B)x + (2A - B - 2C) \\ &\equiv 2x^2 + 3 \end{aligned}$$

$$\Rightarrow \begin{cases} -2A = 2 & \Rightarrow A = -1 \\ -2A - 2B = 0 & \Rightarrow B = 1 \\ 2A - B - 2C = 3 & \Rightarrow C = -3 \end{cases} \Rightarrow y_p(x) = -x^2 + x - 3$$

Thus the general solution

$$y(x) = y_c(x) + y_p(x) = c_1 e^{2x} + c_2 e^{-x} - x^2 + x - 3$$